বল্টু location পাবনায় পড়ানোর capacity নিয়ে transfer হলো। Access করলো স্কুলের performance ছাত্রদের physical type ও char. দেখে organize করলো।

Chapter: 04

Cache Memory

1) Characteristics of Memory System:

Q: ~~Describe~~ Mention key characteristics of memory system

Ans: Key characteristics of computer memory systems:-

(i) **Location:** Refers to wether memory is internal or external.

- Processor
- Internal (main)
- External (secondary)

(ii) **Capacity:** For internal memory, expressed in terms of bytes or words. External memory is typically expressed by bytes.

- word size
- number of words.

(iii) **Unit of Transfer:**

- word
- block

For internal, its equal to the num. of data lines.

(iv) Access Method: This includes the following-

— Sequential Access
— Direct "
— Random "
— Associative "

~~(2020): Explain Access~~

(v) Performance: The performance parameters are used -

— Access time
— cycle time
— Transfer Rate

(vi) Physical Type: The most common:

— Semiconductor
— Magnetic
— Optical
— Magneto-Optical

(v) Physical characteristics:

- Volatile/Nonvolatile
- Erasable/Nonerasable

(vi) Organization: For modern access memory, It's a key design issue. Physical arrangement of bits to form words.

(2) Explain memory access method in computer system.

Ans: Its a key characteristic of of memory system. The Memory Access Method includes the following:

(i) Sequential Access: Memory is organized into units of data, called records. Access must be made in a specific linear sequence. Access time is variable.

(ii) Direct Access: As with sequential access, direct access involves a shared read-write mechanism. Access time is variable. Individual blocks or records have a unique address based on physical location.

(iii) Random Access: Each addressable location in memory has a unique addressing mechanism. Any location can be selected at random and directly accessed and addressed. Main memory and some cache systems are random access.

(iv) Associative: A word is retrieved based on a portion of its contents rather than its address. cache memory may employ it.

(3) Differences between sequential access, direct access and random access.
(Follow No. 02)

(4) Three Performance parameters:-

(i) Access time (Latency): For random access memory, this is the time it takes to perform a read or write operation.

(ii) Memory cycle Time: Consists of the access time plus any additional time required before a second access can commence.

(iii) Transfer Rate: This is the rate at which data ~~wh~~ can be transferred into or out of a memory unit. For random access memory it is equal to 1/cycle time.

For non random access,

$$T_N = T_A + \frac{N}{R}$$

N — Number of bits

R — Transfer rate in bits/s

T_N — Average time to R/W N bits

T_A — Average access time

(5) Relationship among access time, memory cost and capacity? ——— 04.

<u>Ans:</u> There is a trade-off among the three key characteristics of memory namely cost, capacity and access time. At any given time, a variety of technologies are used to implement memory system. The following relationships hold:

i) Faster access time, greater cost per bit.

ii) Greater capacity, smaller cost per bit.

iii) Greater capacity, slower access time.

The designer would like to use memory technologies that provide for large capacity memory. To meet performance requirements needs to use expensive, lower capacity with short access times. The way out of this by memory heirarchy.

(i) Decreasing cost/bit

(ii) Increasing capacity

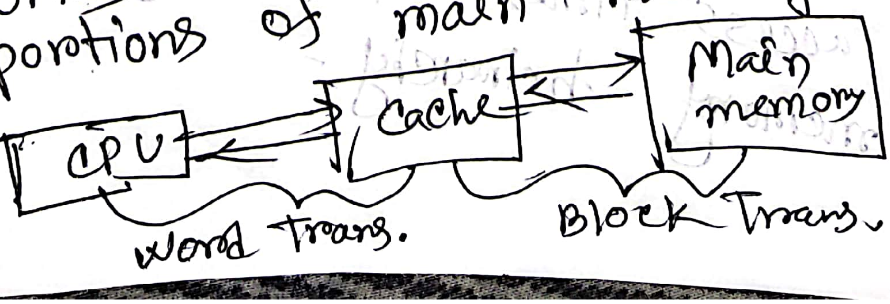
(iii) Increasing access time

(iv) Decreasing frequency of ~~me~~ access of the memory by the processor.

(6) What is cache memory? Why need cache memory?

Ans: Cache memory is a special very high speed memory. ~~to~~ that is used to speed up and synchronizing with high speed CPU.

Cache memory is intended to give memory speed approaching that of the fastest memory available, Provide a large memory size at the price of less expensive type of semiconductor memories. The cache contains a copy of portions of main memory.



(7) Elements of Cache Design:

(i) Cache size
(ii) Mapping function
(iii) Replacement Algorithm
(iv) Write Policy
(v) Line size
(vi) Number of caches.

Cache size:
Be small enough, large enough

(8) Mapping Function:

Que: Difference between direct mapping, associative mapping, set associative mapping.

Ans: Three techniques can be used in mapping.

Direct Mapping: The simplest technique known as direct mapping. ~~The mapping function is implemented using the address.~~ Each block from main memory has only one possible place in the cache organization.

Associative Mapping: Permits each main memory block to be loaded into any line of the cache.

Set Associative Mapping: It is the combination of advantages of both direct and associative mapping. The cache is divided into a number of sets of cache lines; each main memory block can be mapped into any line in a particular set.

(9) What is Replacement Algorithm? Why replacement algorithm necessary?

Ans: When a new block is brought into the cache, one of the existing blocks must be replaced.

For the associative and set associative techniques, a replacement algorithm is needed. To achieve high speed, such an algorithm must be implemented in hardware.

ABCO24, EFCDEF

(10) ~~How does the principal of locality relate to the use of multiple memory levels?~~

(10) For a direct mapped cache, a main memory address is viewed as consisting of three fields. List and define the three fields.

Ans: One field identifies a unique word or byte within a block of main memory. The remaining two fields specify one of the block of main memory. Its identify one of the lines of a cache, and a tag field.

2 fields?

(11) ,, ,, associative ,, ,,

Ans: A tag field uniquely identifies a block of main memory. A word field identifies a unique word or byte ------ 3?

(12) ,, ,, set-associative ---

Ans: same as 10.

12) What is distinction between spatial locality and temporal locality?

Ans: Spatial locality refers to the tendency of execution to involve a number of memory location that are clustered. Temporal locality refers to the tendency for a processor ~~that~~ ~~to~~ access memory locations that have been used recently.

13) Tag, Word, Line calculation:

(i) For the hexadecimal main memory addresses 111111, 666666, BBBBBB, show the following info ....

~~Ans:~~ নিয়ম: 1) ~~convert Hex Value~~ to Binary

2)

8 - 14 - 2
1 43 2

Ans:

[illegible]

[illegible]

1 1 1 1 1 1
1

0001 0001 0001 0001 0001 0001 0.001 0001

Chapter-05

1) Difference between DRAM and SRAM:-

RAM technology is dieded into two technologies dynamic and static. A dynamic RAM is made with cells that store data as charge on capacitors. A static RAM is a digita device using the same logic elements used in the processor. In DRAM, the presence/absence of charge on a capacitor is intepreted as binary 1 or 0. In SRAM, binary values are stored using traditional flip-flop logic gate configurations.

A dynamic memory cell is simpler and smaller than a static memory cell. A DRAM is more dense and less expensive than a corresponding SRAM. DRAMs tend to be favored for large memory requirements. SAMs are generally faster than DRAMs. SRAM is used for cache memory, DRAM is used for main memory.

2) <u>ROM:</u> Read only memory contains a permanent pattern of data that cannot be changed. It is nonvolatile no power source is required to maintain the bit values in memory.

- PROM (Programmable ROM)
- EPROM (Erasable " ROM)
- EEPROM (Electrically erasable)
- Flash memory

why need . . .

disgusting

(3) Explain error correction mechanism of computer system — 4

Ans: বইয়ের page 148

(4) Why need error correction mechanism of computer system?

Ans: same . . .

Chapter-06

(1) What is RAID? Describe RAID. — 1+3

Ans:
RAID or Redundant Array of Independent Disks is a standardized scheme for multiple-disk database design. The RAID scheme consists of seven levels, zero through six.

The levels share three common characteristics:

(i) RAID is a set of physical disk drives viewed by the operating system as a single logical drive.

(ii) Data are distributed across the physical drives of an array.

(iii) Redundent disk capacity is used to store parity information.

The RAID strategy replaces large-capacity disk drives with multiple smaller capacity drives and distributes data in such a way as to enable simultaneous access to data from multiple drives. Use of multiple devices increases the probability of failure.

~~Chapter-07~~

② Mention the key feautures of RAID 0 - RAID 6 —

Page 176 — Table 6.3

Chapter-7

① External Devices — Keyboard/Monitor, disk drive

② I/O Modules:

(i) Breifly describe I/O module with its functionality —— 3

(ii) Why I/O module is used in computer system. Describe its functions —— 2+4

Ans:

The major functions or requirements for an I/O module :-

i) Control and Timing.
ii) Processor Communication.
iii) Device Communication
iv) Data Buffering
v) Error detection

description from page — ~~201~~ মিম আপুর নোট থেকে

An I/O module transfers data from external devices to CPU and vice versa. It contains internal Buffer for temporarily holding these data until they can be sent on.

③ The techniques to Perform I/O:

Q1 — ~~Do~~ criticize programmed I/O and interupt driven I/O

Q2 — Distinguish between program driven and interup driven I/O.

(i) Programmed I/O: With programmed I/O, data are exchanged between the processor and I/O module. The processor executes a program that gives it direct control of I/O operation, including read or righ command, transferring the data. If the processor is faster than the I/O module, this is the wasteful of processor time.

(ii) Interrupt driven I/O: ~~The process~~ issues an I/O command. The processor has to wait a long time for the I/O module to be ready for transmission of data

The process issues on I/O comman, continues to execute other instruc. and is interrupted by I/O module when the later is completed, it is work.

⊞ Other Unsolved Questions: (2018, 2017)

(i) Mention the computer generation with its corresponding technology (from slide or assignment)

(ii) Explain the distinction between the written sequence and time sequence of for branch instruction.

Ans: The operation of a computer, in executing a program, consists of sequence of instruction cycles, with one machine instruction per cycle. This sequence of instruction cycle is not necessarily the same as the written sequence of instructions that make up the program, because of the existence of branching instructions.

The actual execution of instructions follows a time sequence of instructions.

(iii) ~~Differentiate between CISC and RISC architecture. What are their their typical characteristics?~~

Ans: ~~মিস আমার সিটে আছে বাবারা!~~

(iv) Enlist different addressing modes

i) Implicit Addressing Mode
ii) Stack "
iii) Immediate "
iv) Direct "
v) Indirect "
vi) Register Direct "
vii) " Indirect "
viii) Relative "
ix) Indexed "
ix) Base register "
x) Auto increment
xi) Auto decreament

(v) Shortly explain peripheral devices with classification

—— বই - 213 page

(vi) Control signals among system module —

—— মিস আছে কিছুর বোঝে।

(vii) ~~Advantage~~ and disadvantage

(vii) Relation between instruction and micro operations

—— শীট মিস আছে

(vii) What is super capacitor?

সিলেবাসে আছে কিনা জানি নাই ।।